

Solving the Identity-Privacy Paradox: How to Identify Friend or Foe Without Self-Disclosure?

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Abstract. Unmanned aerial vehicles (UAVs) operating in contested federated environments must make identification-friend-or-foe (IFF) decisions without relying on centralised infrastructure or revealing sensitive operational information. We present $\mathcal{R}_{\text{IFF}}$, a non-interactive zero-knowledge attestation protocol that enables a UAV to prove federation membership, possession of an enrolled hardware-bound secret, and policy authorisation without disclosing its identity, role, sector, or command chain. We evaluate a proof-of-concept implementation across four operational scenarios on x86-64 and ARM platforms under controlled network impairment. On x86-64, authentication introduces operationally acceptable overhead, while prover performance on ARM remains the primary deployment bottleneck. We further identify relay resistance as a fundamental limitation of the non-interactive design.

Keywords: zero-knowledge proofs · UAV authentication · identification friend-or-foe · privacy-preserving protocols · federated systems

1 Introduction

Modern UAV operations often rely on commercial-off-the-shelf (COTS) platforms in contested environments, where they face authentication challenges that existing commercial frameworks are not designed to solve. Classical identification-friend-or-foe (IFF) systems rely on shared secrets or centralised certificate infrastructure, but multi-national UAV swarm deployments for defence and humanitarian assistance missions usually offer neither reliable connectivity to ground infrastructure nor the simplicity of a single trusted authority. A UAV encountering an unknown airframe at operational speed must decide in seconds whether to treat it as a coalition asset, locally, without querying a ground control station, and without learning anything about the approaching object’s identity or mission that it does not need to know. The verifying party needs enough assurance to act, but the proving party must not disclose the sensitive operational attributes that would normally provide it: certificate-based authentication fails this requirement because every exchange carries a persistent identifier, allowing

an adversary observing coalition radio-frequency (RF) traffic to link authentication events to specific assets, reconstruct patrol routes, and infer command structures without breaking any cryptographic primitive. In a NATO eastern-flank context, where adversaries conduct long-term passive collection of RF emissions for later analysis [24], this is not a theoretical concern.

To prove that an asset is authorised and non-compromised without disclosing sensitive mission information, we propose a non-interactive zero-knowledge (NIZK) attestation protocol, $\mathcal{R}_{\text{IFF}}$, that encodes three simultaneous trust conditions in a single proof: the prover is a registered member of the federation, its hardware key is consistent with the enrolled reference, and its sector and role credentials are valid under the active policy. The public statement reveals only Merkle roots and a per-session nonce; it reveals no identity, affiliation, trajectory, or command chain.

We make three **contributions**:

- **Formal relation and security analysis.** We define the $\mathcal{R}_{\text{IFF}}$ relation formally and establish five security properties against the principal attack classes applicable to federated UAV mission scenarios.
- **Joint circuit construction.** We introduce a circuit that enforces the enrolled hardware reference across the membership and hardware statements, preventing an adversary who holds partial credential material from two different assets from constructing a valid proof — a threat specific to coalition environments where credential custodianship is distributed across national authorities with heterogeneous security postures.
- **Cross-platform empirical evaluation.** We are unaware of prior work jointly demonstrating a hardware-bound NIZK IFF relation with a cross-platform evaluation covering baseline performance, network-impairment sensitivity, adversarial security, and a deployability assessment translating measured latency into minimum encounter distances. $\mathcal{R}_{\text{IFF}}$ should be read as a deployment methodology for privacy-preserving coalition IFF rather than a new NIZK primitive.

The protocol relation uses only collision-resistant hash functions and Merkle path verification, with no elliptic curve or pairing assumptions embedded, an important property given current post-quantum transition recommendations [8]. Section 8 discusses why migratability is a property of the relation; the deployed Groth16 protocol does not share it.

2 Related Work

UAV authentication has been studied from several angles, none of which fully addresses the combination of requirements imposed by contested multi-national swarm operations. Existing approaches each address a piece of this problem without solving it as a whole: none lets a drone convince a stranger in the air that it belongs to the same coalition while telling that stranger nothing else, and without either party contacting the ground. Identity-based schemes assume

a ground station the drone already shares a key with. Hardware-rooted schemes tie a credential to a specific piece of silicon. Flight-privacy schemes conceal where a drone has been. The four groups below correspond to the columns of Table 1.

Identity-based authentication. Most schemes treat UAVs as Internet-of-Things (IoT) devices authenticated via public-key infrastructure (PKI) or symmetric keys [17], providing mutual authentication and session key establishment but tying trust to stable identifiers. Protocols based on elliptic-curve cryptography (ECC), such as LAPEC [25], use short-lived pseudonyms to limit passive linkability. Yet, the ground station still maps long-term identities to pseudonyms, creating a central dependency. All these schemes require infrastructure connectivity.

Hardware-rooted authentication. Schemes based on physical unclonable functions (PUFs) bind identity to device-specific silicon variations, removing the need for stored long-term secrets [21]. Petzi et al. [19] combine PUF responses with ZK proofs to authenticate IoT devices against strong physical adversaries, representing the closest prior work to the hardware binding considered here. Their interactive ECC-based ZK proof does not support non-interactive forwarding or multi-observer verification.

Zero-knowledge attestation. ZAPS [18] uses zero-knowledge succinct non-interactive arguments of knowledge (zk-SNARKs) to prove flight path compliance without revealing waypoints, and Koulianos et al. [15] prove location authorisation via a public blockchain. Koulianos et al.’s on-chain verification is a platform feature of the blockchain rather than a protocol-level forwarding mechanism, and the proofs remain bound to a specific smart contract instance; neither scheme designs the proof for forwarding to an arbitrary third-party observer, which is why Table 1 marks both “No” under Forwarding despite the underlying NIZK proofs being publicly verifiable by anyone holding the verification key. Walshe et al. [23] propose NIZK authentication for IoT in low-connectivity settings using pre-computed Merkle paths for single-message authentication, but omit hardware attestation. We are unaware of an existing UAV NIZK scheme that enforces cross-statement binding across membership, hardware, and policy within a single circuit.

Post-quantum cryptography. Prior work integrates CRYSTALS/Kyber and CRYSTALS/Dilithium into UAV authentication [20], showing post-quantum cryptography (PQC) is feasible on constrained platforms. Boschini et al. [2] construct lattice-based post-quantum SNARKs for the Ring Short Integer Solution (RSIS) and Ring Learning With Errors (RLWE) problems, showing practical post-quantum NIZK proofs of possession; our work is designed to be compatible with such backends as tooling matures.

This work addresses the lack of a protocol that jointly supports decentralised operation, hardware-bound attestation, cross-statement binding under partial credential compromise, multi-observer forwarding, and a relation structure compatible with post-quantum backends. Table 1 summarises this gap; Section 7.2 revisits these schemes with measured performance figures.

Table 1. Gap analysis against closest prior work. ZK: zero-knowledge construction. HW-bound: hardware-bound secret possession. Forward.: protocol-level support for forwarding a proof to an arbitrary third-party observer, as distinct from mere public verifiability by a party already holding the verification key (all listed NIZK/SNARK schemes are publicly verifiable in this weaker sense). Relay-res.: relay/proximity resistance. PQ-migr.: relation structure compatible with post-quantum backends.

Scheme	ZK	HW-bound	Forward.	Relay-res.	PQ-migr.
LAPEC [25]	No	No	No	No	No
Petzi et al. [19]	Yes	Yes	No	No	No
ZAPS [18]	Yes	No	No	No	No
Koulianos et al. [15]	Yes	No	No*	No	No
Walshe et al. [23]	Yes	No	No	No	Unclear
This work	Yes	Yes	Yes	No	Yes

*Koulianos et al. [15] verify proofs on an Ethereum smart contract, giving any blockchain observer a verification path; we do not count this as protocol-level forwarding since it is a platform feature of the blockchain rather than a mechanism of the proof system itself, and the proofs are bound to a specific contract instance.

3 Preliminaries

Table 2 summarises the notation used throughout the rest of the paper. $\mathcal{R}_{\text{IFF}}$ is built from three standard ingredients: a collision-resistant hash function, a sparse Merkle tree, and a NIZK proof system in the common reference string (CRS) model, whose generic definitions (completeness, soundness, zero-knowledge, and the negligible-probability bound $\mu(\lambda)$ used throughout this paper) are standard and are given in Appendix A. What follows is the protocol-specific detail needed to read Section 4.

Table 2. Notation used in the $\mathcal{R}_{\text{IFF}}$ specification.

Symbol	Meaning
φ	Public input vector $(\varphi_{\text{reg}}, \varphi_{\text{nonce}}, \varphi_{\text{epoch}}, \varphi_{\text{hw}}, \varphi_{\text{pol}})$
w	Private witness vector $(w_{\text{reg}}, w_{\text{hw}}, w_{\text{pol}})$
π	NIZK proof
$H(\cdot)$	Collision-resistant hash function (Poseidon over BN254 scalar field)
k_{hw}	Live hardware-derived key (volatile memory only)
$h_{\text{hw}}^{\text{enrol}}$	$H(k_{\text{hw}}^{\text{enrol}})$, stored in OD enrolment registry
ℓ	Registry leaf: $H(H(ID_{\text{priv}}) h_{\text{hw}}^{\text{enrol}})$
VerifyMembership	Merkle path verification against federation registry root
VerifyAuthorisation	Merkle path verification against policy tree root

Informally, each ingredient does a specific job. The Merkle tree lets a member prove its entry is on the federation’s list of enrolled assets: the whole list is hashed into a tree, and a short chain of sibling hashes is enough for a verifier holding only the root to check membership. The NIZK proof then lets the prover convince the verifier that it genuinely holds secrets satisfying that condition, without

revealing what those secrets are or how it knows them; because it is a single self-contained message, it can be checked offline and passed on to other observers. The hardware-rooted secret ties the whole proof to one specific piece of physical hardware — a device-specific key that cannot be cloned onto another platform and never sits in storage in the clear, so a valid proof means this particular airframe, not just any holder of the credential.

Registry instantiation. A sparse Merkle tree (SMT) over key space $\{0, 1\}^{256}$ assigns each possible key a unique leaf position and assigns empty leaves predetermined default hash values (formal definition in Appendix A). This supports efficient proofs of membership and non-membership, and the root hash depends only on the set contents and not on insertion order. Revocation is implemented by removing the corresponding leaf and distributing the updated root.

Proof system. This work uses Groth16 [11], which produces the smallest known pairing-based NIZK proofs: exactly three elliptic curve group elements⁴ (137 bytes over BN254) with constant-size verification. In the CRS model [1] $\text{CRS} = (\text{pk}, \text{vk})$ is generated in a one-time trusted setup; the prover computes $\pi \leftarrow \text{Prove}(\text{pk}, x, w)$ offline, and any party holding vk can check $\text{Verify}(\text{vk}, x, \pi)$ independently and deterministically, enabling multi-observer verification without runtime contact with the prover. The relation is compiled to a Rank-1 Constraint System (R1CS) and then a Quadratic Arithmetic Program (QAP); in-circuit hashing uses Poseidon [10], defined natively over prime fields and substantially reducing constraint count relative to SHA-256.

Hardware-rooted secrets and physical unclonable functions. The protocol requires only that the device can derive a stable, device-specific secret k_{hw} that is hard to extract or clone and that does not need to be stored persistently in plaintext; it is otherwise agnostic to the rooting mechanism, and a trusted platform module (TPM), trusted execution environment (TEE), secure element (SE), or eSIM could each serve as a drop-in alternative to the construction below. We instantiate k_{hw} with a physical unclonable function (PUF), which exploits sub-micron manufacturing variations to implement a device-specific challenge-response mapping, because it additionally avoids storing the secret persistently even in protected hardware. At enrolment, a weak PUF with a fixed challenge space is evaluated to derive k_{hw} , and the trusted authority stores $h_{\text{hw}}^{\text{enrol}} = H(k_{\text{hw}}^{\text{enrol}})$ while the raw key is erased. At runtime, the device regenerates k_{hw} from the PUF and a fuzzy extractor [7] corrects noise via stored helper data. Because the key exists only transiently in volatile memory during use, PUFs provide stronger hardware binding than storage-based alternatives against adversaries with temporary physical access [19].

Operational context. UAVs are organised into swarms coordinating under a federated mission authority that pre-distributes trust anchors via the NATO Federated Mission Networking (FMN) model [22] before deployment. Once in

⁴ Three group elements over BN254 are 128 bytes in compressed form. The 137-byte figure quoted throughout is the size of the serialised proof our prototype emits, and is the figure Appendix E is based on; the nine-byte difference is libsnark’s serialisation framing.

the field, UAVs operate without real-time ground control station connectivity. Cryptographic interaction is initiated only after an approaching object is classified as potentially friendly by an external sensing layer (e.g. EO/IR tracking, radar, ADS-B, or operator cueing); target acquisition and classification are out of scope of $\mathcal{R}_{\text{IFF}}$ and are treated as a black-box input supplying the candidate identity hypothesis used during verification. The outcome of the protocol is a binary local trust classification and does not constitute an engagement or kinetic decision.

4 The $\mathcal{R}_{\text{IFF}}$ Protocol

The notation introduced in Table 2 (Section 3) is used throughout this section.

4.1 Design Rationale

The protocol jointly verifies three conditions in a single interaction: federation membership, hardware-key consistency, and policy authorisation, a layered structure conceptually similar to multi-factor document verification schemes such as electronic passport control [5].

We use a joint circuit instead of three independent proofs because, in a coalition federation, credential material is distributed across national custodians with heterogeneous security postures: an adversary who separately compromises asset A’s identity credential and asset B’s hardware key could otherwise submit independent membership and hardware proofs for A and B that each verify correctly without revealing they originate from different enrolled assets. The joint circuit closes this by enforcing the enrolled hardware reference $h_{\text{hw}}^{\text{enrol}}$ as a shared wire constrained simultaneously by the registry leaf computation and the hardware equality check, so the R1CS is unsatisfiable for any pair $(ID_{\text{priv}}^A, k_{\text{hw}}^B)$ not appearing as a leaf in the federation registry.

This binding is a witness-consistency property and not a novel primitive, equivalent to hashing a combined leaf $H(ID \parallel h_{\text{hw}})$ in a conventional credential system, but the three-statement decomposition creates an opportunity for it to be lost if the statements are composed carelessly, a risk specific to multi-custodian coalition credential issuance; we prove it formally in Theorem 1 (Section 5.2).

4.2 Assumptions

H is collision-resistant and efficiently computable in a circuit, and the NIZK proof system satisfies completeness, knowledge soundness, and zero knowledge, with the CRS generated honestly or via a multi-party computation with at least one honest participant. The observer drone (OD) maintains authenticated local copies of $\text{Registry}_{\text{root}}^*$, $\text{Policy}_{\text{root}}^*$, and the hardware enrolment registry, updated only via authenticated out-of-band channels, and samples φ_{nonce} uniformly at

random per encounter without reuse. The device contains a PUF producing sufficiently high-entropy responses; k_{hw} is produced on-device in volatile memory, never written to persistent storage, and not subject to physical extraction before revocation (compromised devices are instead revoked through registry updates). In the prototype evaluation, hardware key derivation is simulated; the protocol structure does not depend on this choice, but real-PUF evaluation (intra-chip variability, bit-error rate, fuzzy-extractor failure rate, temperature/aging sensitivity) is left to future work.

Trust Boundary. The ZKP establishes that the prover knows witnesses satisfying the relation with respect to the public inputs; it does not guarantee freshness at the time of attestation and does not protect against side channels, communication metadata, or relay (wormhole) attacks, which are out of scope of the relation itself and treated as a fundamental limitation in Section 5.3.

4.3 Public and Private Inputs

The OD constructs the partial public input vector $\{\varphi_{\text{reg}}, \varphi_{\text{nonce}}, \varphi_{\text{epoch}}, \varphi_{\text{pol}}\}$ and transmits it to the approaching object (AO). The AO appends $\varphi_{\text{hw}} = H(H(k_{\text{hw}}) \parallel \varphi_{\text{nonce}})$, computed from its live hardware key and the received nonce, to form the complete public input vector $\varphi = \{\varphi_{\text{reg}}, \varphi_{\text{nonce}}, \varphi_{\text{epoch}}, \varphi_{\text{hw}}, \varphi_{\text{pol}}\}$.

The private witness $w = \{w_{\text{reg}}, w_{\text{hw}}, w_{\text{pol}}\}$ is never transmitted. $w_{\text{reg}} = (ID_{\text{priv}}, \text{Path}_{\text{reg}})$ is the private identity and its Merkle authentication path; the registry leaf $\ell = H(H(ID_{\text{priv}}) \parallel h_{\text{hw}}^{\text{enrol}})$ binds both the identity and the enrolled hardware reference to the same leaf, ensuring S1 and S2 are proven for the same enrolled asset within the circuit. $w_{\text{hw}} = (k_{\text{hw}})$ is the hardware-derived key produced on-device at attestation time; the circuit checks $H(H(k_{\text{hw}}) \parallel \varphi_{\text{nonce}}) = \varphi_{\text{hw}}$ and enforces $H(k_{\text{hw}}) = h_{\text{hw}}^{\text{enrol}}$, binding the live key to the enrolled reference committed in the leaf of S1; this witness is never stored persistently. $w_{\text{pol}} = (\text{Sector}_{\text{AO}}, \text{Role}_{\text{AO}}, \text{Path}_{\text{pol}})$ contains the attribute values and their Merkle path to φ_{pol} ; the attribute tuple is bound to the active epoch inside the circuit.

4.4 The NIZK Relation

$$\begin{aligned} R_{\text{IFF}} = \{(\phi, w) \mid & \text{Joint_Valid}(w_{\text{reg}}, w_{\text{hw}}, \phi_{\text{reg}}) \\ & \wedge \text{HW_Valid}(w_{\text{hw}}, \phi_{\text{nonce}}, \phi_{\text{hw}}) \\ & \wedge \text{Policy_Valid}(w_{\text{pol}}, \phi_{\text{epoch}}, \phi_{\text{pol}}) \} \end{aligned} \quad (1)$$

All three conjuncts must hold simultaneously. **Joint_Valid** reflects that membership and hardware binding are verified jointly against a single registry leaf, ensuring both are proven for the same enrolled asset.

Statement 1 (Membership and Hardware Binding). The prover demonstrates that its identity and hardware reference are jointly enrolled in the federation registry:

$$\text{VerifyMembership}(H(H(ID_{\text{priv}}) \parallel h_{\text{hw}}^{\text{enrol}}), \text{Path}_{\text{reg}}, \varphi_{\text{reg}}) = \text{True}$$

where $h_{\text{hw}}^{\text{enrol}}$ is treated as an internal circuit wire whose value is enforced by Statement 2 to equal $H(k_{\text{hw}})$. Revocation removes the leaf from the registry, simultaneously invalidating both the identity and hardware binding of the revoked asset.

Remark 1. The leaf encodes $h_{\text{hw}}^{\text{enrol}} = H(k_{\text{hw}}^{\text{enrol}})$ rather than the raw key, so a compromise of the registry tree does not directly expose the underlying hardware secret.

Statement 2 (Enrolment-Bound Hardware Secret). The circuit enforces two constraints jointly:

$$H(H(k_{\text{hw}}) \parallel \phi_{\text{nonce}}) \stackrel{?}{=} \phi_{\text{hw}} \quad \text{and} \quad H(k_{\text{hw}}) \stackrel{?}{=} h_{\text{hw}}^{\text{enrol}}$$

The first constraint binds the hardware key to the current session via the nonce. The second enforces that k_{hw} is consistent with the enrolled reference committed in the registry leaf of Statement 1, closing the cross-asset substitution attack of Section 4.1. The OD independently verifies $\varphi_{\text{hw}} \stackrel{?}{=} H(h_{\text{hw}}^{\text{enrol}} \parallel \varphi_{\text{nonce}})$ as an anchor check before invoking the verifier.

Remark 2. Statement 2 proves possession of a hardware-derived secret consistent with the enrolled commitment: it does not attest hardware integrity, untampered firmware, secure boot, runtime state, or a trusted execution environment, nor that k_{hw} was freshly derived from the physical PUF at attestation time. We term this *enrolment-bound secret possession* and reserve *hardware integrity attestation* for the strictly stronger notion this protocol does not provide; freshness and integrity are platform-level properties outside the relation, following the separation of concerns in [19]. The OD’s plaintext anchor check above is itself computable without the in-circuit constraint, since the OD already holds $h_{\text{hw}}^{\text{enrol}}$ and φ_{nonce} ; what Statement 2 adds is not this check but the binding of $h_{\text{hw}}^{\text{enrol}}$ as a wire shared with Statement 1’s leaf computation, so that the same value is proven consistent with both the registry membership and the live hardware key within a single witness (Theorem 1).

Statement 3 (Policy Authorisation). The prover demonstrates authorisation under the active policy epoch:

$$\text{VerifyAuthorisation}(H(\text{Sector}_{\text{AO}} \parallel \text{Role}_{\text{AO}} \parallel \varphi_{\text{epoch}}^{\text{cred}}), \text{Path}_{\text{pol}}, \varphi_{\text{pol}}) = \text{True}$$

where the circuit additionally enforces $\varphi_{\text{epoch}}^{\text{cred}} \in \{\varphi_{\text{epoch}} - 1, \varphi_{\text{epoch}}\}$. The one-epoch tolerance accommodates delayed credential propagation while rejecting credentials older than one epoch.

4.5 Protocol Workflow

During pre-deployment, the FMN distributes CRS, vk, registry anchors, and policy anchors to all observers. During an encounter, the OD samples φ_{nonce}

uniformly at random and transmits the partial public input vector to the AO. The AO generates a session-bound NIZK proof: it derives k_{hw} from its PUF, computes φ_{hw} , constructs the witness vector, evaluates $\mathcal{R}_{\text{IFF}}(\varphi, w)$ locally, generates $\pi \leftarrow \text{Prove}(\text{pk}, \varphi, w)$, erases k_{hw} , and transmits (φ, π) to the OD. The OD performs lightweight deterministic anchor checks (root matching, nonce cache lookup, epoch validation, hardware commitment check) before invoking $\text{Verify}(\text{vk}, \varphi, \pi)$. Any failure returns `unknown`; only a verified proof returns `trusted`. An accepted tuple is disseminated to neighbouring observers for independent re-verification.

Because the proof is non-interactive and bound to φ , it can be forwarded between observers without recomputation by the prover. A forwarded tuple (π, φ) verifies correctly at any secondary observer holding authenticated local copies of $\text{Registry}_{\text{root}}^*$, $\text{Policy}_{\text{root}}^*$, the hardware enrolment registry, and the pinned vk , by determinism of Verify : all such observers reach identical trust decisions on the same (π, φ) without coordination or contacting the original prover, and a single completing verification cycle is therefore valid for the full swarm rather than only the OD that ran it.

This forwarding mechanism only propagates the result of Statements 1–3; it cannot attest that the forwarding observer is reporting on the airframe currently in its sensor range instead of relaying a proof generated elsewhere. This limitation is treated formally in Section 5.3.

5 Security Analysis

Before stating the properties $\mathcal{R}_{\text{IFF}}$ satisfies, we make explicit what is required to satisfy them, and against which adversary.

Adversary model. $\mathcal{A}$ is a PPT algorithm holding vk and the public anchors, with full control of the wireless layer: it observes every transcript and may inject, drop, reorder, modify and replay messages. It has two oracles. $O_{\text{cor}}(d, p)$ returns one part $p \in \{\text{id}, \text{hw}, \text{pol}\}$ of enrolled device d ’s credential material, modelling the multi-custodian issuance of Section 4.1; write $\text{Cor}(d)$ for the parts obtained for d . $O_{\text{prv}}(d, \varphi)$ returns an honest proof by d on statement φ . $\mathcal{A}$ neither influences Setup nor alters an honest OD’s pinned anchors or nonce cache (Section 4.2). In the challenge session, the OD samples a fresh φ_{nonce}^* , and $\mathcal{A}$ may not query O_{prv} on any statement containing it. That last restriction is exactly the relay capability of Section 5.3, excluded because no mechanism in the design prevents it; writing it as an oracle restriction rather than leaving it implicit is what makes SR1–SR5 and SR7 provable statements, and makes the excluded capability impossible to overlook. Fig. 1 (Appendix F) shows the corresponding adversarial positions.

Table 3 lists seven security requirements the protocol was designed against. Each is stated below as a property quantified over all PPT $\mathcal{A}$ in the model above, and each is discharged by a specific claim, theorem, or inherited proof-system property established in the remainder of this section.

- **SR1 (Replay resistance).** For every PPT $\mathcal{A}$: (a) a proof generated on a statement carrying $\varphi_{\text{nonce}}^{(1)}$ is accepted on a statement carrying $\varphi_{\text{nonce}}^{(2)} \neq \varphi_{\text{nonce}}^{(1)}$

with probability at most $\mu(\lambda)$; and (b) a tuple already accepted by an OD is not accepted again by that OD. The per-verifier scope of (b) is deliberate and is not sufficient at swarm level; see Section 8.

- **SR2 (Session unlinkability)**. For every PPT $\mathcal{A}$ that does not hold the hardware enrolment registry, given two challenge transcripts each run with one of two enrolled devices of $\mathcal{A}$ ’s choosing, $\mathcal{A}$ decides whether the same device produced both with advantage at most $\mu(\lambda)$ over guessing. The exclusion of registry holders is necessary (Claim 3).
- **SR3 (Soundness under partial compromise)**. For every PPT $\mathcal{A}$ obtaining $\text{Cor}(A) = \{\text{id}, \text{pol}\}$ and $\text{Cor}(B) = \{\text{hw}\}$ for enrolled $A \neq B$, an honest OD outputs trusted in the challenge session with probability at most $\mu(\lambda)$.
- **SR4 (Hardware binding)**. If an honest OD outputs trusted on (φ, π) then, except with probability at most $\mu(\lambda)$, the extracted witness contains a k_{hw} with $H(k_{\text{hw}}) = h_{\text{hw}}^{\text{enrol}}$ for the very leaf proved under Statement 1, and φ and π reveal k_{hw} to no party.
- **SR5 (Context binding)**. If an honest OD outputs trusted on (φ, π) then, except with probability at most $\mu(\lambda)$, φ_{reg} and φ_{pol} equal that OD’s pinned anchors and the credential epoch in the witness lies in $\{\varphi_{\text{epoch}} - 1, \varphi_{\text{epoch}}\}$.
- **SR6 (Relation-level post-quantum readiness)**. The relation itself must use only collision-resistant hashing and Merkle path verification, with no pairing-based assumption embedded, so that it can be re-instantiated with a post-quantum backend without redefining the relation. Unlike SR1–SR5 and SR7 this is a structural requirement on the relation rather than a game against $\mathcal{A}$; it is discharged by inspection of Section 4.4.
- **SR7 (Zero-knowledge privacy)**. For every PPT $\mathcal{A}$ holding only $(\varphi, \pi, \text{vk})$ there is a simulator producing transcripts $\mathcal{A}$ distinguishes from real ones with advantage at most $\mu(\lambda)$; in particular ID_{priv} , k_{hw} , $\text{Sector}_{\text{AO}}$ and Role_{AO} are hidden from every party, registry holders included.

Two requirements from the design brief are deliberately *not* listed above because $\mathcal{R}_{\text{IFF}}$ does not satisfy them, and we do not want Table 3 to imply otherwise: proximity/relay resistance and full (cross-federation) anonymity. Both are discussed as open limitations in Sections 5.3 and 8.

5.1 Inherited Properties

$\mathcal{R}_{\text{IFF}}$ inherits completeness, knowledge soundness, and zero knowledge from Groth16 under the q -Power Knowledge of Exponent and q -Strong Diffie-Hellman assumptions in the generic group model [11]. Knowledge soundness gives a PPT extractor that, from an accepting proof, recovers a single witness assignment satisfying all three statements jointly; since $\mathcal{R}_{\text{IFF}}$ is a single R1CS, this assignment cannot mix credential material from different transcripts. We use this extraction property directly in Theorem 1.

Zero knowledge hides ID_{priv} , k_{hw} , $\text{Sector}_{\text{AO}}$, and Role_{AO} from any party holding only $(\varphi, \pi, \text{vk})$, and session unlinkability follows from φ_{nonce} and φ_{hw} being fresh per session. This does *not* extend to anonymity against any party holding

Table 3. Security requirements, the mechanism that establishes each, the test that exercises it, and the corresponding MITRE ATT&CK for Industrial Control Systems (ICS) technique.

Requirement	Established by	Test	Technique
SR1 – Replay resistance	Claim 1 (Nonce-Bound Replay Protection), §5.2	T2, T3	T0856, T0830
SR2 – Session unlinkability	Claim 3 (Attribute-Hiding), §5.2, for parties without the enrolment registry; zero knowledge with fresh $\varphi_{\text{nonce}}/\varphi_{\text{hw}}$, §5.1	T1	T0842
SR3 – Soundness under partial compromise	Theorem 1 (Asset Binding), §5.2; confirmed empirically by T6, §7.1	T6	T0849
SR4 – Hardware binding	Statement 2, §4.4; Theorem 1, §5.2	T6	T0849
SR5 – Context binding	Claim 2 (Context Binding), §5.2	T4, T5	T0830, T0856
SR6 – Relation-level readiness	PQ Relation structure, §4.4; discussed in §8	–	–
SR7 – Zero-knowledge privacy	Inherited zero-knowledge property, §5.1	–	T0842
Excluded	Relay/proximity resistance (§5.3) and cross-federation anonymity (Claim 3): no mitigating mechanism by design	–	T0830

the hardware enrolment registry, which includes every observer and not only the interrogating one (Claim 3). The protocol therefore hides the prover’s attributes from all parties, but hides its identity only from parties outside the federation. We term this *attribute-hiding attestation*, a weaker guarantee than anonymous authentication.

5.2 Protocol-Specific Properties

Claim 1 (Nonce-Bound Replay Protection). A proof π generated under $\varphi_{\text{nonce}}^{(1)}$ cannot be accepted by a verifier presenting $\varphi_{\text{nonce}}^{(2)} \neq \varphi_{\text{nonce}}^{(1)}$, except with probability at most $\mu(\lambda)$.

Proof (Sketch). Replay protection follows from nonce freshness (the OD samples φ_{nonce} uniformly per encounter and never reuses a value, Section 4.2), stateful nonce-cache tracking at the OD, and the binding of $\varphi_{\text{hw}} = H(H(k_{\text{hw}}) \parallel \varphi_{\text{nonce}})$ into the statement φ itself, so a proof generated under one nonce is, by Groth16 statement-soundness, never accepting against a statement carrying a different nonce. Collision resistance of H closes the residual gap of an adversary finding a different k'_{hw} that reproduces the same φ_{hw} under a fresh nonce.

Claim 2 (Context Binding). A proof valid under epoch φ_{epoch} and roots $\text{Registry}_{\text{root}}$, $\text{Policy}_{\text{root}}$ cannot satisfy $\mathcal{R}_{\text{IFF}}$ under a different epoch or different roots without a recomputed witness, except with probability at most $\mu(\lambda)$.

Proof (Sketch). The in-circuit constraint $\varphi_{\text{epoch}}^{\text{cred}} \in \{\varphi_{\text{epoch}} - 1, \varphi_{\text{epoch}}\}$ and Groth16 statement-soundness rule out acceptance under a mismatched epoch; given correct OD-side anchor pinning, satisfying a different $\text{Policy}_{\text{root}}$ requires either control of the issuing mission authority or breaking collision resistance of H .

Claim 3 (Attribute-Hiding Within a Shared Federation). Under a shared $\text{Registry}_{\text{root}}$ and $\text{Policy}_{\text{root}}$, no party without the hardware enrolment registry can determine the issuing authority, national affiliation, or command chain of the prover from φ or π , except with probability at most $\mu(\lambda)$.

Proof (Sketch). This follows from zero knowledge (π reveals nothing about w beyond $\mathcal{R}_{\text{IFF}}(\varphi, w) = 1$) together with the architectural choice of a single shared root with no per-nation partitioning; per-nation anchors would restore state-level attribution. The restriction to parties without the enrolment registry is necessary. Section 4.5 requires a secondary observer to hold that registry in order to verify a forwarded tuple, and any registry holder can compute $H(h_{\text{hw}}^{\text{enrol}} \parallel \varphi_{\text{nonce}})$ for each enrolled asset and match it against the public φ_{hw} , recovering the prover’s identity at cost linear in registry size. The claim therefore gives identity privacy against outsiders and attribute privacy against insiders. Full anonymity is not achieved, and this is inherent: the OD’s anchor check on Statement 2 is that same computation.

Theorem 1 (Asset Binding). Assume Groth16 knowledge soundness for $\mathcal{R}_{\text{IFF}}$ [11] and collision resistance of H . Then no PPT adversary, given enrolled credential material for two distinct assets $A \neq B$ (specifically ID_{priv}^A and k_{hw}^B), can produce an accepting proof π for the joint *riff* circuit under any statement φ , except with probability at most $\mu(\lambda)$, where μ is negligible in λ and under the stated assumptions may be conservatively bounded by $O(2^{-\lambda})$.

Proof. Suppose for contradiction that an adversary $\mathcal{A}$, possessing only ID_{priv}^A and k_{hw}^B for $A \neq B$, produces an accepting (φ, π) that the OD relies on as evidence of asset A ’s authorisation. By Groth16 knowledge soundness, a PPT extractor $\mathcal{E}$, run against $\mathcal{A}$, recovers, except with probability at most $\mu(\lambda)$, a witness w satisfying $\mathcal{R}_{\text{IFF}}(\varphi, w) = 1$. Because $\mathcal{R}_{\text{IFF}}$ is one conjunctive relation compiled to a single R1CS, the value $h_{\text{hw}}^{\text{enrol}}$ used in w ’s registry-leaf check $\ell = H(H(ID_{\text{priv}}^A) \parallel h_{\text{hw}}^{\text{enrol}})$ and the value constrained by Statement 2’s equality $H(k_{\text{hw}}) \stackrel{?}{=} h_{\text{hw}}^{\text{enrol}}$ must be identical in w . By construction of enrolment, the leaf ℓ^A that the OD accepts as belonging to asset A was inserted as $\ell^A = H(H(ID_{\text{priv}}^A) \parallel H(k_{\text{hw}}^A))$ for A ’s own enrolled key k_{hw}^A . For w to satisfy *VerifyMembership* against ℓ^A while $\mathcal{A}$ supplies only k_{hw}^B , the extracted witness’s hardware-key component must equal some k'_{hw} with $H(k'_{\text{hw}}) = H(k_{\text{hw}}^A)$; since $\mathcal{A}$ possesses only $k_{\text{hw}}^B \neq k_{\text{hw}}^A$ (independently sampled at enrolment), $\mathcal{A}$ has thereby found $k'_{\text{hw}} \neq k_{\text{hw}}^A$ with $H(k'_{\text{hw}}) = H(k_{\text{hw}}^A)$ – a collision in H , contradicting our assumption that H is collision-resistant. Hence, except with probability at most $\mu(\lambda)$, $\mathcal{A}$ cannot produce a proof that verifies against A ’s registry leaf using k_{hw}^B .

This is confirmed empirically in Section 7.1 (T6): a witness combining one enrolled asset’s identity with a different asset’s hardware key is mechanically constructible, *libsark* does not check R1CS satisfiability before proving, but is rejected by the joint-circuit verifier, while the same two individually valid proofs are each separately accepted under the modular *all* mode, which enforces consistency only via OD-side anchor checks (Section 6) and carries no analogous guarantee.

Theorem 1 discharges SR3 and SR4.

5.3 Relay Attacks: Fundamental Limitation of NIZKs

Because (π, φ) is a non-interactive bearer token, an adversary positioned between the AO and OD can forward a legitimate, freshly generated proof without holding any enrolled credentials: a relay drone E in radio range of both a friendly drone F and the OD passes the OD’s nonce challenge to F , obtains F ’s honestly generated proof, and forwards it, causing the OD to accept it as evidence that the airframe in its sensor range is friendly. The nonce establishes session freshness; it says nothing about physical proximity. Hardware binding does not close this gap either, since k_{hw} is never revealed in any form the OD could use to fingerprint the transmitting radio.

$\mathcal{R}_{\text{IFF}}$ *authenticates possession of a valid, enrolment-consistent, currently authorised credential. The spatial proximity or physical presence of the transmitting platform is a separate question the proof does not answer.* A deployment without a proximity-binding mechanism is vulnerable to this relay scenario, an instance of the wormhole attack on wireless networks [13]. The limitation is structural to non-interactive bearer-token designs generally, not specific to our construction: any scheme accepting a single self-contained (φ, π) tuple is forwardable by definition. Until added, $\mathcal{R}_{\text{IFF}}$ is a credential verification layer, meant to be composed with a separate proximity assurance mechanism. On its own, it is not a complete IFF solution. Closing the gap requires mechanisms outside the current design, distance bounding [3,12,14], channel binding, or proof-of-proximity, each reintroducing interactivity the design avoids; we leave this to future work.

6 Implementation

The prototype instantiates the three statements of $\mathcal{R}_{\text{IFF}}$ in C++ using the libsnark Groth16 implementation [16] over BN254 (ALT_BN128) with Poseidon hashing and depth-4 sparse Merkle trees⁵, orchestrated by a Python harness over a message bus that injects controlled network conditions across the four flight scenarios. libsnark provides direct, stable access to constraint-level R1CS introspection, which we rely on for the per-source constraint breakdown of Table 4 and the depth-scaling measurements of Section 8, and for constructing the deliberately inconsistent witness used in T6 (Section 7.1); Section 8 discusses the resulting evaluation gap against more recent proving stacks.

A modular `all` mode proves Statements 1–3 as separate circuits with consistency enforced only by OD-side anchor checks; we retain it for ablation but do not use it in the evaluation, as (Theorem 1) it does not provide the asset-binding guarantee of the joint `riff` circuit.

⁵ Source code and reproducibility instructions available at <https://github.com/emasujster/RIFF>.

Table 4. Circuit complexity by statement (top; proof size is constant at 137 bytes for all Groth16 instances over BN254, regardless of circuit complexity) and constraint breakdown of the joint $\mathcal{R}_{\text{IFF}}$ circuit at depth 4 by source (bottom). S1–S3 are instantiated with the same Poseidon and Merkle-path gadgets as the joint circuit, so their constraint counts sum exactly to the joint total.

Statement	Constraints	Public inputs	Proof (B)	Pub. input file (B)
S1 Identity	3,462	1	137	78
S2 Hardware	1,152	2	137	120
S3 Policy	3,463	2	137	82
Joint $\mathcal{R}_{\text{IFF}}$ (depth 4)	8,077	5	137	276

Constraint source	Constraints	Share
Poseidon: identity path ($H(ID_{\text{priv}})$, registry leaf)	1,150	14.2%
Poseidon: hardware binding ($H(k_{\text{hw}})$, session commitment)	1,150	14.2%
Poseidon: policy path (attribute hash, policy leaf)	1,150	14.2%
Merkle registry path (4 levels)	2,312	28.6%
Merkle policy path (4 levels)	2,312	28.6%
Equality constraints (direct R1CS, Statement 2)	3	<0.1%
Total	8,077	100%

6.1 Circuit Implementation

In-circuit hashing uses Poseidon [10] over the BN254 scalar field (8 full rounds, 31 partial rounds, $\alpha = 17$). Statements 1, 3, and the joint circuit enforce Merkle membership via a binary tree gadget; Statement 2 enforces two Poseidon hash equalities directly. The joint circuit shares $h_{\text{hw}}^{\text{enrol}}$ as a wire across Statements 1 and 2, collapsing one duplicated constraint. Table 4 summarises circuit complexity at the prototype’s native depth-4 registry and decomposes the joint circuit’s 8,077 constraints by source.

Each Merkle level contributes one Poseidon evaluation per tree, so the two Merkle-path rows dominate the circuit at this depth; Section 8 reports how this composition shifts as registry depth grows. Appendix E details how the resulting 137-byte proof and 276-byte public input vector map onto MAVLink v2 frame limits for transport over the autopilot data link.

6.2 Experimental Setup

All x86-64 experiments were conducted on a single host running the Windows Subsystem for Linux 2 (WSL2) (AMD Ryzen 7 4800H, 8 cores, Linux kernel 6.6.114). ARM experiments used a Raspberry Pi 3B+ running Ubuntu 24.04 (ARM Cortex-A53, 1 GB RAM). Both platforms use ALT_BN128 over GNU Multiple Precision (GMP) arithmetic. Network impairment was injected at the Authentication Bridge level via per-message Gaussian delay and probabilistic drop, isolating protocol-level latency sensitivity from simulator-specific networking artefacts. Four scenario profiles were defined (Table 10, Appendix D) to span distinct encounter geometries; each (scenario, network-condition) combination was run for $n = 80$ trials at baseline, $n = 50$ per loss level, and $n = 30$ (x86-64) or $n = 140$ (ARM) per delay level. The cross-platform comparison of Sections

7-7.3 uses the prototype’s native registry scale of 16 enrolled assets (depth-4 Merkle tree); registry-depth scaling is measured separately in Section 8.

7 Evaluation

On x86-64, prover wall time is stable at approximately 963 ms across all scenarios ($\sigma < 25$ ms; 95% confidence interval (CI) for Proximate Hold: [959.7, 965.9] ms), confirming that proving cost is determined by circuit complexity and is independent of flight geometry. The verifier completes in approximately 10.8 ms, a ratio of 90:1. This asymmetry is operationally significant: the OD, which bears the verification load for all encountered assets, incurs negligible per-encounter cryptographic cost, enabling a single observer node to validate multiple concurrent sessions without becoming compute-bound. End-to-end latency ranges from a median of 1,829 ms (Proximate Hold) to 2,121 ms (Mission Separation).

On ARM (Raspberry Pi 3B+), prover wall time averages 13,970–14,341 ms, a 14.5–14.8 $\times$ slowdown attributable to GMP’s generic arithmetic backend in the absence of assembly-optimised BN254 support; verifier wall time averages approximately 58 ms (5.4 $\times$). The prover-to-verifier asymmetry is preserved at 243:1, so the principle of assigning verification to the higher-capability node remains valid even on embedded hardware. Assembly-optimised Cortex-A pairing arithmetic exists in other zk-SNARK projects and would plausibly close much of this gap; the 5–15 $\times$ figure used in Section 7.3 is illustrative. We did not measure it for this codebase.

7.1 Adversarial Security Tests

Six adversarial tests were conducted against the protocol’s security claims, each exercising a distinct attack vector from the threat model of Table 9 or, for T6, directly probing Theorem 1; all six returned the expected outcome on the evaluated platform (Table 8, Appendix B; Table 5 below). T1–T5 are targeted functional tests of specific mechanisms (nonce caching, proof-byte tampering, verification-key pinning, epoch checking, and forwarding correctness); T6 is a constructive adversarial test of cross-asset witness mixing against the joint circuit. None of this constitutes a systematic adversarial cryptanalysis. We did not test, and identify as evaluation gaps, malicious-prover behaviour beyond simple tampering and witness mixing, CRS-compromise scenarios, PUF noise/error-correction edge cases, revocation-lag windows, denial-of-service against the OD’s verification path, registry-root desynchronisation across observers, Byzantine-observer behaviour, and stale-forwarding replay across long time windows (Section 8).

T1–T5 validated forwarding correctness, nonce-bound replay protection, proof-tampering resistance, cross-verification-key impersonation resistance, and epoch context binding; all five behaved as expected, with rejections occurring at the lightweight anchor pre-check before pairing computation wherever the threat permitted it.

Table 5. T6: cross-asset witness-mixing test. Steps 1–3 establish that each credential is individually valid and that the modular **all** mode cannot detect the cross-asset mismatch; step 4 confirms the joint **riff** circuit rejects it.

Step	Mode	What it shows	Result
1	identity (seed 0)	Asset A’s identity alone is valid	PASS
2	hardware (seed 1)	Asset B’s hardware alone is valid	PASS
3	all (seeds 0 / 1)	Both accepted independently; all mode has no cross-asset binding	PASS
4	riff_mixed (id 0, hw 1)	Joint circuit rejects mismatched witness	FAIL (rejected)

T6: cross-asset witness mixing. To empirically validate Theorem 1, we constructed a witness for the joint circuit combining the identity material of one enrolled asset ($\text{id_seed} = 0$) with the hardware key of a different enrolled asset ($\text{hw_seed} = 1$). Table 5 reports the four-step result. Steps 1–2 confirm each asset’s credential is independently valid in isolation. Step 3 confirms that the modular **all** mode, which checks Statements 1–3 as separate circuits, accepts both proofs independently and has no mechanism to detect that they originate from different enrolled assets, empirically reproducing the gap described in Section 6. Step 4 is the critical test: the same mismatched credential pair, submitted as a single witness to the joint **riff** circuit, produces a proof that the verifier rejects (exit code 1). Because libsnark does not check R1CS satisfiability before proving, witness construction and proof generation in step 4 succeed mechanically; the rejection occurs only at verification, consistent with Theorem 1’s argument that no satisfying assignment exists for mismatched credential material except with probability at most $\mu(\lambda)$ under collision resistance of H .

7.2 Comparison Against Prior Work

Table 6 extends the gap analysis of Table 1 with measured proof sizes and verifier latency where comparable figures are available. $\mathcal{R}_{\text{IFF}}$ is the only scheme that states plainly that it lacks relay resistance; the others leave the question unaddressed. Performance figures are specific to our Groth16/libsnark instantiation and do not extend to more recent SNARK toolchains, which we have not benchmarked.

7.3 Deployability Analysis

The Transit Encounter scenario is the most operationally informative: two UAVs closing at 13.7 m/s remain within 230 m of each other for approximately 33 s, but authentication must complete during the approach phase before closest approach, which lasts approximately 16.8 s. On x86-64, a single cycle completes in under 2.2 s, fitting eight complete authentication attempts within that window; on ARM, a single cycle takes 25–28 s, so zero cycles complete before closest approach. This only addresses whether a proof *can be computed and verified in time*; it says nothing about whether the resulting trust decision is bound to the correct physical airframe, the separate relay question of Section 5.3.

Table 6. Comparison against prior work. Proof sizes exclude public inputs. “Attr.-hiding” denotes the attribute-hiding property of Section 5.1, not full anonymity. “PQ-migr.” denotes relation-level migratability (Section 8), not protocol-level post-quantum security as instantiated.

Scheme	Proof	Attr.-hiding	HW-bound*	Forward.	Relay-res.	PQ-migr.
ZAPS [18]	128 B	Partial	No	No	No	No
Koulianos et al. [15]	–	No	No	No	No	No
Petzi et al. [19]	–	No	Yes (PUF)	No	No	No
Walshe et al. [23]	–	Partial	No	No	No	Unclear
This work	137 B	Yes	Sim. (PUF)	Yes	No	Yes

*“HW-bound” denotes possession of an enrolment-bound hardware secret (Remark 2), not hardware integrity attestation. Verification latency for ZAPS (~ 3 – 5 ms, indicative estimate, not reported by the authors) and proof size/latency for Koulianos et al., Petzi et al., and Walshe et al. are omitted where the cited work does not report a directly comparable Groth16-class NIZK construction.

The critical separation distance $D_{\text{crit}} = v \cdot T_{\text{cycle}}$ gives the minimum starting separation at which at least one full cycle completes before two platforms closing at speed v meet. D_{crit} stays under 110 m on x86-64 across the evaluated speed range (5–40 m/s), versus ~ 351 m on ARM at the Transit Encounter baseline. Since D_{crit} is linear in T_{cycle} , any cut in ARM prover time reduces D_{crit} proportionally at every closure speed. ARM deployability is an engineering problem with a known lever to pull; nothing here points to a hard ceiling (cf. the unverified 5–15 $\times$ estimate of Section 7).

Network delay adds ~ 3 ms per 1 ms of injected one-way delay on x86-64, consistent with the three-message structure; on ARM this gradient is masked by prover variance ($\sigma \approx 400$ ms), so cutting network delay buys little relative to cutting prover cost (Fig. 2, Appendix G). Packet loss hurts availability. It does not hurt survivor latency. Success rates stay $\geq 96\%$ (x86-64) and $\geq 98\%$ (ARM) at 1% loss, dropping to 86–94% and 88–96% at 5% loss, with survivor latency essentially unchanged. Loss eliminates cycles instead of slowing survivors. Operationally, the 6–14% of x86-64 cycles that time out at 5% loss would misclassify friendly assets as unverified, so doctrine must set retry logic and human-in-the-loop thresholds to bound this fratricide risk.

8 Discussion

8.1 Limitations

This subsection collects every limitation in one place, ordered by how much each constrains deployment; Section 8.2 onward returns to the forward-looking material.

Relay resistance as the primary open limitation. As established in Section 5.3, $\mathcal{R}_{\text{IFF}}$ authenticates credential possession. That does not establish physical proximity, and the protocol remains vulnerable to relay by a cooperating or compromised intermediary. This is the most consequential limitation for the IFF use case; we do not regard the protocol as deployment-ready for adversarial IFF use without a proximity-binding mechanism.

Replay to a secondary observer. The nonce cache in the protocol of Section 4.5 is maintained per-verifier, and the freshness check rejects only a nonce that *that* verifier has already seen; it does not check whether that verifier is the one who issued it. A secondary observer that did not itself challenge the AO therefore has an empty cache entry for that nonce, so a captured, still-valid tuple (φ, π) presented to a fresh observer within the same epoch passes the freshness check and verifies. This is distinct from the real-time relay attack above: no low-latency forwarding is required, and it is a deferred rather than live replay, but it shares the same root cause: the forwarding model of Section 4.5 gives a secondary observer no way to distinguish a legitimately forwarded proof from an adversary-replayed one. Closing it requires either shared nonce state across observers or the same class of proximity/channel binding already identified as future work for relay resistance.

Evaluation scope. All experiments were run on a single host per platform, with software-simulated telemetry, software-injected network impairment, and a simulated (not physical) PUF. The results thus characterise protocol logic and cryptographic cost; RF propagation, embedded scheduling effects, interrupt latency, and PUF error-correction overhead sit outside what was measured. Delay and packet loss were also injected independently; their combined effect is untested, and the loss model draws independent per-message drops instead of the bursty loss typical of real wireless channels, so the reported loss-survival rates (Section 7.3) are a best-case estimate. The same caveat applies to the ARM comparison in Section 7: our setup cannot separate the software arithmetic backend from the Cortex-A53 hardware itself, so the measured slowdown is an upper bound on what the chip alone would impose.

No machine-checked verification. The claims of Section 5.2 are analytical, backed by the adversarial tests of Section 7.1, but not by a symbolic model in a tool such as ProVerif, Tamarin, or Scyther. Given the compact three-message structure of the protocol, a minimal model of the nonce- and epoch-binding claims (SR1, SR5) is feasible and is identified as near-term future work. The adversarial tests of Section 7.1 are structural and deterministic. They carry no statistical weight, so they do not substitute for a symbolic model.

8.2 Interpretation and Outlook

Post-quantum migratability is a property of the relation, not the protocol. $\mathcal{R}_{\text{IFF}}$ uses only collision-resistant hashing and Merkle verification, with no pairing assumptions embedded, and is migratable in that narrow sense. But the deployed protocol’s practical properties, 137-byte proofs, 10.8 ms verification, two-packet MAVLink bundling, the D_{crit} figures, are consequences of the Groth16 instantiation. It also requires a relation-specific CRS generated from a trapdoor; for a multi-national coalition, the multi-party setup ceremony [6] raises political and operational questions alike — onboarding new nations, member withdrawal, emergency revocation, and CRS rotation, questions this work assumes are resolved by an honestly run ceremony rather than answering. Separately,

the deployed BN254 curve offers roughly 100 bits of security and is not NIST-approved; BLS12-381 (~ 128 bits) is a drop-in alternative that would raise the margin without changing the relation, at the cost of regenerating the CRS and keys. A STARK-based [4] or lattice-based [2] backend would avoid the trusted setup entirely but would plausibly move proof sizes into the tens-to-hundreds-of-kilobyte range, breaking two-packet MAVLink bundling and requiring the deployability analysis to be redone.

Registry scale. We measured constraint counts, QAP size, and prover wall time on x86-64 for the joint circuit at depths 4, 8, 10, and 12 (16 to 4,096 enrolled assets), reported in Table 7. Constraints grow linearly at +578 per level, but prover wall time grows more slowly: the QAP size only doubles when the constraint count crosses a power-of-two boundary, so most depth increases add Poseidon evaluations without changing QAP cost. At depth 12 (4,096 assets), proving remains under 2 s on x86-64.

Table 7. Measured circuit scaling with registry depth (x86-64). Constraints grow linearly; prover time grows more slowly due to QAP power-of-two boundaries.

Depth	Leaves	Constraints	QAP size	Prove (ms)
4	16	8,077	8,192	1,013
8	256	12,701	16,384	1,522
10	1,024	15,013	16,384	1,660
12	4,096	17,325	32,768	1,927

Proving-stack currency. $\mathcal{R}_{\text{IFF}}$ was evaluated on libsnark, a comparatively dated library; a modern toolchain (arkworks, gnark, halo2, SP1/Risc0) would materially change the absolute timings of Section 7, particularly on ARM.

Design trade-offs. Merkle roots were used instead of anonymous credential schemes for fast revocation and constrained-device efficiency, at the cost of making federation membership and the active policy epoch visible in the public inputs to any channel observer. Hardware-derived secrets were prioritised over full federation-wide anonymity, which is why the OD must already hold a candidate identity hypothesis from non-cryptographic sensing before the hardware check can run (Section 3, Section 5.1). This provides attribute privacy; it falls short of anonymity. The non-interactive design that enables multi-hop forwarding also means proof timing and bundle transmission remain observable as traffic metadata even though their content is hidden.

Broader applicability. Proving multiple simultaneous claims without disclosing the underlying attributes extends beyond UAVs to the EU Digital Identity Wallet, anonymous credential systems, and decentralised identity infrastructures; the attribute-hiding-versus-anonymity distinction of Section 5.1 applies wherever the verifier already holds a pre-existing identity hypothesis about the party it engages.

9 Conclusion

Two UAVs meeting in the field, with no ground-station connectivity, need to decide in seconds whether the other is a coalition asset, without either side learning anything about the other that it does not need to know; that is the operational problem this paper set out to solve. This paper presented $\mathcal{R}_{\text{IFF}}$, a NIZK attestation protocol for UAV identification-friend-or-foe decisions in contested multi-national environments, encoding federation membership, enrolment-bound hardware secret possession, and policy authorisation as a single Groth16 proof that hides all sensitive prover attributes beyond the interrogating observer’s prior identity hypothesis. The joint circuit’s cross-asset binding guarantee, established formally and confirmed empirically (Theorem 1, T6), prevents credential composition under partial compromise, unlike the modular `all` mode. On x86-64, it completes in under 2.2s with a 137-byte proof, passing six adversarial tests; on ARM, prover slowdown imposes encounter-distance constraints via $D_{\text{crit}} = v \cdot T_{\text{cycle}}$. Relay resistance remains the most significant open problem. The protocol proves that a credential is valid and enrolment-consistent. It does not establish that the credential is being transmitted by the airframe currently in sensor range, so a well-placed relay can still forward someone else’s legitimate proof as its own. Bridging that gap needs a proximity-binding mechanism that reintroduces some of the interactivity the design otherwise avoids (Section 8). Distance/channel-binding, realistic-scale evaluation, modern toolchain benchmarking, and a broader adversarial campaign are left to future work.

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A Standard Cryptographic Definitions

This appendix collects the generic, textbook-level definitions referenced from Section 3: readers already familiar with NIZK proofs and standard security-parameter conventions can skip directly to Section 4.

Cryptographic hash functions. A cryptographic hash function $H : \{0, 1\}^* \rightarrow \{0, 1\}^n$ is collision-resistant if no probabilistic polynomial-time (PPT) adversary can find (x, x') with $x \neq x'$ such that $H(x) = H(x')$.

Sparse Merkle trees. A sparse Merkle tree (SMT) over key space $\{0, 1\}^{256}$ assigns each possible key a unique leaf position and assigns empty leaves predetermined default hash values. This supports efficient proofs of membership and non-membership, and the root hash depends only on the set contents and not on insertion order.

NIZK proofs. Let $\mathcal{R} \subseteq \mathcal{X} \times \mathcal{W}$ be an NP relation with public statement x and private witness w . A proof system for $\mathcal{R}$ satisfies completeness, knowledge soundness (a computationally bounded adversary cannot produce an accepting proof without a valid witness), and zero-knowledge (a simulator without access to w produces transcripts indistinguishable from real proofs) [9]. In the common reference string (CRS) model [1], $\text{CRS} = (\text{pk}, \text{vk})$ is generated in a one-time trusted setup; the prover computes $\pi \leftarrow \text{Prove}(\text{pk}, x, w)$ offline, and any party holding vk can check $\text{Verify}(\text{vk}, x, \pi)$ independently and deterministically, enabling multi-observer verification without runtime contact with the prover.

Security parameter and negligible probability. Let λ denote the security parameter, already implicit in expressions such as $ID_{\text{priv}} \xleftarrow{\$} \{0, 1\}^\lambda$. A function $\mu(\lambda)$ is *negligible* if, for every polynomial $p(\cdot)$, there exists λ_0 such that $\mu(\lambda) < 1/p(\lambda)$ for all $\lambda > \lambda_0$. Throughout this paper, adversaries are PPT algorithms, and security claims hold except with probability at most $\mu(\lambda)$; under the assumed collision resistance of H and the soundness guarantees of the instantiated proof system, these negligible advantages may be conservatively upper-bounded by $O(2^{-\lambda})$.

B Adversarial Test Outcomes (T1–T5)

The constructive cross-asset witness-mixing test (T6) is reported separately in Table 5 in the main body.

Table 8. Adversarial security test outcomes (functional mechanism tests; they do not amount to a comprehensive adversarial evaluation; see Section 7.1).

ID	Attack	Result
T1	Secondary observer re-verification (forwarded bundle verified independently)	PASS
T2	Nonce replay detection (proof for nonce_A rejected with nonce_B)	PASS
T3	Proof tampering detection (corrupted bytes rejected)	PASS
T4	Cross-VK impersonation (attacker proof rejected under legitimate VK)	PASS
T5	Epoch confusion (epoch=3 proof rejected with epoch=5)	PASS

C Threat Model Attack Mapping

The underlying mechanisms are argued formally as Claims in Section 5.2.

Table 9. Considered attacks and protocol mitigations, mapped to MITRE ATT&CK for Industrial Control Systems (ICS). Relay/wormhole attacks (Section 5.3) are deliberately excluded, as the protocol does not mitigate them.

Attack	Technique	Mitigation
Credential replay	T0856, T0830	Nonce freshness; stateful nonce cache
Compromised drone impersonation	T0849	Knowledge soundness; revocation
Attribute disclosure to eavesdroppers	T0842	Zero knowledge; session unlinkability
Authority and federation inference	T0842	Shared federation root; ZK property
Context confusion	T0830, T0856	Epoch binding; policy root anchor

D Flight Scenario Profiles

Table 10. Flight scenario profiles. Base delay arises from the distance model at zero injected delay.

Scenario	Description	Mean dist. Base delay	
Proximate Hold	Both UAVs stationary at 4 m	4 m	0 ms
Transit Encounter	UAV1 passes at 13.7 m/s; separation 0–230 m	107 m	54 ms
Mission Separation	Both UAVs diverging to 300 m	160 m	79 ms
Joint Operation	Phased mission with approach, split, loiter	117 m	64 ms

E MAVLink Transmission

MAVLink v2, the standard autopilot data-link protocol, constrains the application payload to 253 bytes per frame. The Groth16 proof π is 137 bytes; φ occupies 276 bytes in libsark’s decimal text format. The minimal bundle is 413 bytes raw (552 in Base64), fitting exactly 2 MAVLink v2 packets; a packed 32-byte binary encoding reduces this to $137 + 5 \times 32 = 297$ bytes, also within 2 packets. This analysis is specific to the Groth16 instantiation: a post-quantum proof system (Section 8) would change these figures substantially.

F Threat Model Diagram

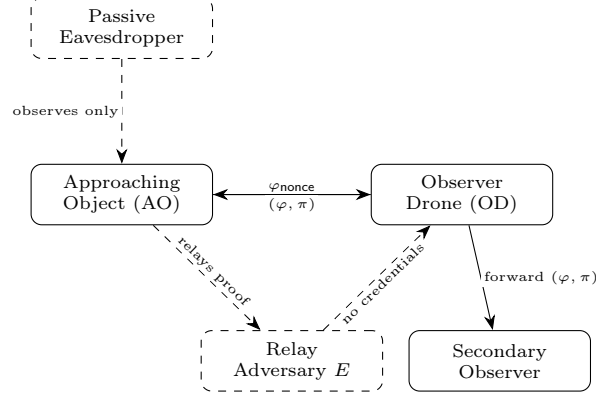


Fig. 1. Threat-model actors. Solid arrows: legitimate protocol flow (challenge, proof, forwarding to a secondary observer). Dashed arrows: adversarial positions in scope (passive eavesdropper, Section 5.1) and out of scope of the relation itself (relay adversary E , Section 5.3).

G Network Impairment Results

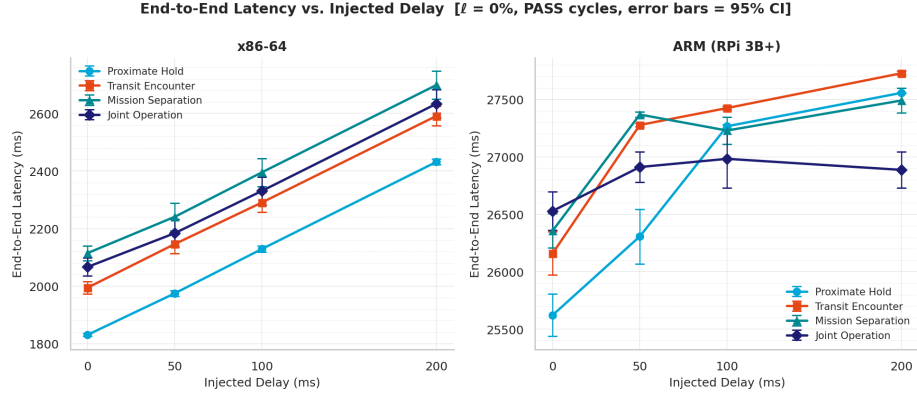


Fig. 2. End-to-end latency vs. injected one-way delay on x86-64 (PASS cycles only), across all four scenarios at zero packet loss. The relationship is strictly linear (≈ 3 ms per 1 ms of injected delay) across all scenarios, giving $L_{e2e} \approx L_{\text{base}} + 3d$. At 1% per-message loss, success rates remain $\geq 96\%$ (x86-64) and $\geq 98\%$ (ARM); at 5% loss they degrade to 86–94% and 88–96% respectively. Packet loss eliminates cycles instead of slowing survivors. Error bars show 95% CI of the mean.